QA TEST REPORT · PRODUCTION

# Howler Quality Assurance Audit

A full-surface QA pass across authentication, moderation, messaging, rate limiting, accessibility, and security — executed with real browser automation and live API calls against the production deployment.

- **135** test cases
- 19 categories
- Environment: **Production** (howler-teal.vercel.app)
- Executed 2026-07-18

## Results at a glance

Of 135 cases, 119 were actually executed; 16 are Blocked or Not Run for documented reasons (a second test account, out-of-scope test imagery, or infrastructure access this engagement didn't have — see the appendix).

| 90 | 29 |
|---|---|
| Pass 66.7% | Fail 21.5% |
| **6** | **10** |
| Blocked 4.4% | Not run 7.4% |

Of the **119** cases actually executed, **75.6%** passed. Two additional defects (**NEW-01**, **NEW-02**) were discovered during testing and are included in the totals above.

# Methodology

This was a real-environment audit, not a synthetic test suite run against mocks — every result below reflects an actual request against production.

### Real browser automation

Playwright drove actual Chromium, Firefox, and WebKit sessions through the real UI — clicks, form fills, uploads — not simulated events.

### Live API validation

Every endpoint was also called directly with an authenticated session to check status codes, payloads, and error handling beneath the UI layer.

### Resilience & adversarial cases

Network throttling, forced offline states, delayed/failed responses, tampered cookies, and malformed payloads were used to probe failure paths, not just happy paths.

### Guardrails on real users

Every action visible to a real person, billed against a real API, or targeting another user's content (new DMs, reports, image analysis) was confirmed case-by-case before it ran, and test content was cleaned up immediately after.

**Explicitly out of scope this round:** five AI Image Analysis cases (IMG-02–06) require sourcing real adult or violent test imagery, which was not done under any circumstance; two Text Filtration cases were skipped by scope decision; the remaining Blocked/Not Run items need a second test account, a non-UW login, or Supabase dashboard access not available in this engagement.

# Priority findings

Nine issues worth discussing first, ranked by priority. The full list of 29 failures and 6 blocked cases is in the appendix below.

CRITICAL

TXT-02

### Comment profanity filter has a word-specific bypass

The word "shit" passes through the comment filter untouched — reproduced twice — while the identical word is correctly blocked in posts, and a different profane word ("fuck") is correctly blocked in comments too. Likely a context-scoring quirk on short text rather than a total failure, but it's a live, reproducible gap in a shipped moderation feature.

CRITICAL

MSG-03

### Sending a message has no membership check

GET /api/messages verifies the caller is a conversation participant before returning data; POST /api/messages does not — confirmed by source review. Only a foreign-key constraint stood between this test and an unauthorized insert. Given a real conversation ID, any authenticated user could likely message into a conversation they're not part of.

CRITICAL

SEC-06

### Internal database errors reach the client

No credentials or secrets leak, but unhandled exceptions across posts, comments, and other routes return raw Postgres error text — column names, constraint names, type errors — directly in the response body. An internal-schema disclosure worth closing before wider launch.

HIGH

SEC-08

### Session cookie is readable by JavaScript

The auth cookie carries SameSite=Lax (a reasonable CSRF baseline) but httpOnly:false and secure:false. No XSS vector exists today (see SEC-01), but if one is ever introduced the session is trivially stealable — this is in fact exactly how test sessions were captured for this audit.

HIGH

SEC-07

### A corrupted session cookie crashes the server

Replaying a tampered auth cookie against GET /api/auth returns an unhandled 500 crash instead of a clean 401. Any malformed cookie — tampering, corruption, or a buggy client — takes the request down rather than degrading to "unauthenticated."

HIGH

IMG-07 · IMG-09

### Bad image uploads crash the post route — and still cost money

A renamed non-image file or a truncated image both crash posts/route.js with an unhandled 500, because Vision returns a null response that the code reads without a null check. Worse: the request still bills a Vision API call and consumes an upload-quota slot before it fails — a bad actor could burn through both budgets with garbage files.

HIGH

A11Y-01 · A11Y-02

### Core interactions are unreachable by keyboard or screen reader

Like, comment, report, and delete controls are plain s with onClick handlers — no tabindex, no role, no keyboard handler — so they can't be reached with Tab or activated with Enter/Space. The comments modal also lacks role="dialog" and aria-modal, so screen readers won't announce it.

MEDIUM

ERR-04

### /api/reports has no error handling at all

Malformed JSON crashes it with a completely empty 500 — no error message whatsoever. Separately, an empty payload is accepted as a valid report, creating a row with a null post and null reason. Of every input-handling gap found this session, this route needs the most immediate attention.

MEDIUM

RT-03

### The main feed has no realtime updates

A post created in a second session never appeared in an already-open feed without a manual refresh — PostFeed.js has no Supabase realtime subscription at all. Only direct messages actually have realtime, which contradicts the README's claim of "live updates for interactions and posts."

**Also discovered during testing (not in the original plan):**

- NEW-01 — GET /api/posts 500s for a fully unauthenticated caller (a null user ID fails a uuid cast in Postgres). Not reachable through normal navigation today, but a real gap in a public API route.
- NEW-02 — Deleting a post never removes its image from Storage; the file stays publicly reachable indefinitely, accumulating cost and outliving the reason it was deleted.

## Category breakdown

All 19 categories, sorted by failure count — where the problems concentrate, not just where the volume is.

# Full results

All 135 test cases, grouped by category. Search or filter by status — matching sections expand automatically.

135 of 135 test cases shown

## Accessibility

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| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| A11Y-01 | Keyboard-only navigation through the feed | Like/comment/report/delete controls are plain divs with onClick — no tabindex, no role, unreachable by keyboard. | High | Fail |
| A11Y-02 | Screen reader announces modals and key content | Comments modal has no role="dialog" or aria-modal — won't be announced as a dialog. | High | Fail |
| A11Y-03 | Color contrast meets WCAG AA | axe-core found "serious" contrast violations: 16 elements on the feed, 18 inside the comments modal. | Medium | Fail |
| A11Y-04 | Escape key and focus return close modals | Comments modal does not close on Escape (ReportModal does, per source). | Medium | Fail |
| A11Y-05 | Form validation errors are announced accessibly | The category-required error is plain text with no aria-live region — screen reader users aren't notified. | Medium | Fail |
| A11Y-06 | Images have appropriate alt text | Zero image-alt violations found by axe-core. | Low | Pass |

## Security

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| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| SEC-01 | XSS via post content | React's default escaping held — an onerror payload never executed and never rendered as raw HTML. | Critical | Pass |

| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| SEC-02 | SQL/NoSQL injection via query params | Search/profile safely parameterized; but a non-UUID postId on /api/comments leaks a raw Postgres error via 500. | High | Fail |
| SEC-03 | IDOR via guessed/incremented IDs | No successful unauthorized access found across posts/comments/DMs (consolidates POST-06, MSG-02, MSG-03 findings). | Critical | Pass |
| SEC-04 | Rate limiter can't be bypassed via header spoofing | A deliberately sub-threshold batch wasn't large enough to prove or disprove a bypass — avoided further load on production. | Medium | Blocked |
| SEC-05 | File upload blocks path traversal | A malicious filename was ignored entirely — the storage path is built server-side from the user id and timestamp. | Medium | Pass |
| SEC-06 | Secrets/internals never leak in API responses | No credentials leaked, but several routes return raw Postgres error text directly to the client. | Critical | Fail |
| SEC-07 | Tampered session cookie is rejected cleanly | A corrupted cookie crashes the server with an unhandled 500 instead of a clean 401. | High | Fail |
| SEC-08 | CSRF protection on state-changing requests | SameSite=Lax gives a baseline, but the session cookie is httpOnly:false and secure:false — readable via JavaScript. | High | Fail |
| SEC-09 | Reporter identity isn't exposed to the reported user | report_creator_id never appears in the feed response. | High | Pass |

**Posts** 12 / 12

| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| POST-01 | Create a valid text-only post | Created via the real UI flow; appeared correctly in the live feed. | Critical | Pass |
| POST-02 | Post without a category | Succeeded with a null category_id — the "select a category" rule is client-side only, not enforced by the server. | Medium | Fail |
| POST-03 | Feed pagination returns non-overlapping pages | No ID overlap between pages (small current dataset limits deep-pagination coverage). | High | Pass |
| POST-04 | Filter feed by category | Returned only posts matching the requested category. | Medium | Pass |
| POST-05 | User can delete their own post | Deleted via the real UI trigger; confirmed gone from the feed. | High | Pass |
| POST-06 | User cannot delete another user's post | Returned success:true but silently matched zero rows — not actually deleted. Functionally safe, but the misleading response is a UX/API concern. | Critical | Pass |
| POST-07 | Deleting a non-existent post ID | Graceful no-op, success:true, no error. | Low | Pass |
| POST-08 | Deleting a post with no id parameter | Correctly returned 400 'Missing post ID'. | Medium | Pass |
| POST-09 | Like/comment counts on feed items are accurate | Counts matched exactly after a controlled like + 2 comments. | Medium | Pass |
| POST-10 | user_has_liked reflects only the requester's own like | Correct — reflected only the current user's like state. | Medium | Pass |
| NEW-01 | GET /api/posts 500s for a fully unauthenticated caller | A null user id serializes as the string "null" and fails a uuid cast in Postgres. Not reachable through normal navigation, but a real gap in a public route. | Medium | Fail |
| NEW-02 | Deleted posts don't clean up their Storage image | Confirmed: an image stayed publicly reachable (200 OK) minutes after its post was | Medium | Fail |

| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| | | deleted — delete_post never calls storage.remove(). | | |

## Error Handling / Edge Cases

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| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| ERR-01 | 5xx errors show a friendly message, not raw internals | Consolidates several routes that return raw Postgres errors, an empty crash body, or null-reference text directly to the client. | Medium | Fail |
| ERR-02 | Offline handling while posting | Submitting while offline leaves the compose modal stuck on "Posting…" forever — no try/catch around the fetch. | Medium | Fail |
| ERR-03 | Empty states render correctly | A zero-result search returns clean empty arrays for the UI to render an empty state from. | Low | Pass |
| ERR-04 | Malformed JSON body is rejected cleanly | /api/comments leaks a raw parser error via 500; /api/reports has no try/catch at all and crashes with a completely empty 500 body. | Medium | Fail |
| ERR-05 | Back/forward navigation preserves state | URL correctly returned to the prior search state. | Low | Pass |
| ERR-06 | Concurrent delete requests on the same post | Both concurrent deletes returned success:true harmlessly — consistent no-op behavior, not a race condition. | Low | Pass |

## AI Image Analysis

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| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| IMG-01 | Safe image is approved and posted | Benign synthetic image passed SafeSearch and posted normally. | Critical | Pass |
| IMG-02 | VERY_LIKELY adult image is rejected | Not run — requires sourcing real adult-content test imagery, | Critical | Not run |

| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| | | out of scope for this engagement. | | |
| IMG-03 | POSSIBLE adult-content image is still rejected | Not run, same imagery constraint as IMG-02. | High | Not run |
| IMG-04 | LIKELY/VERY_LIKELY violent image is rejected | Not run, same imagery constraint as IMG-02. | Critical | Not run |
| IMG-05 | POSSIBLE-violence-only image is allowed through | Not run, same imagery constraint as IMG-02. | Medium | Not run |
| IMG-06 | Vision's 'racy' field currently isn't checked | Not run; code review shows detections.racy is never read in posts/route.js. | Medium | Not run |
| IMG-07 | Non-image file renamed to .jpg | Unhandled 500 — Vision returns null for non-image data and the route crashes reading .adult. Still billed the Vision call and burned an upload-quota slot. | High | Fail |
| IMG-08 | Very large image (20MB+) | Rejected at the platform layer (413) before reaching app code — the app has no explicit size cap of its own. | Medium | Pass |
| IMG-09 | Corrupted / truncated image file | Same crash as IMG-07 — null safeSearchAnnotation, unhandled 500. | Medium | Fail |
| IMG-10 | Text-only post skips the Vision check | Confirmed across dozens of text-only posts — no Vision call, no quota consumed. | Low | Pass |

## Comments

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| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| COM-01 | Add a top-level comment | Created via the real UI with correct joined profile data. | High | Pass |
| COM-02 | Add a nested reply | Reply correctly persisted with its parent_id. | High | Pass |

| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| COM-03 | Comments are returned in correct order | Returned in strict ascending created_at order. | Medium | Pass |
| COM-04 | GET /api/comments with a malformed postId | A non-UUID postId leaks a raw Postgres error via an unhandled 500 (the missing-param case alone is handled cleanly). | Low | Fail |
| COM-05 | User can delete their own comment | Deleted via the real UI; confirmed removed. | High | Pass |
| COM-06 | User cannot delete another user's comment | Not run — no other user's comment was available this session; POST-06 confirms the identical ownership-check pattern elsewhere. | Critical | Not run |
| COM-07 | Comment on a non-existent post_id | Unhandled 500 with a raw foreign-key constraint error instead of a clean 4xx. | Low | Fail |
| COM-08 | Deep (3-level) reply chains thread correctly | Each reply correctly attributed to its direct parent, no breakage. | Medium | Pass |

## Reports

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| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| RPT-01 | Submit a report with a valid reason | Succeeded via direct API call — the report icon only renders for other users' posts, so self-reporting has no UI path (not a security issue). | High | Pass |
| RPT-02 | Report with no reason selected | Server accepted it anyway (reason stored as null) — the 'select a reason' rule is client-side only. | Medium | Fail |
| RPT-03 | Reporting while logged out | Correctly returned 401. | High | Pass |
| RPT-04 | Same post reported multiple times | No uniqueness constraint — the same user can report the same post repeatedly. | Low | Fail |

| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| RPT-05 | Reported post isn't auto-hidden | Stayed visible after reporting, consistent with a manual-review moderation workflow. | Low | Pass |

## Realtime & Optimistic UI

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| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| RT-01 | Liking updates the UI instantly | Count updated before the network request could have completed. | Medium | Pass |
| RT-02 | Failed like rolls back correctly | Optimistic count correctly reverted after a forced network failure. | High | Pass |
| RT-03 | New posts appear live for other users | Did not appear without a refresh — PostFeed has no realtime subscription at all, contradicting the README's "live updates" claim. | Medium | Fail |
| RT-04 | New posts appear optimistically before the server responds | Not optimistic — the post only appears after the real server response returns; a "Posting…" state is shown instead. | Medium | Fail |
| RT-05 | Realtime reconnects after a dropped connection | A message sent during a simulated outage arrived within 10s of reconnecting. | Medium | Pass |

## Text Filtration

10 / 10

| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| TXT-01 | Explicit profanity in a post is rejected | Correctly blocked with 400. | Critical | Pass |
| TXT-02 | Profanity in a comment is rejected | "shit" bypassed the comment filter twice (reproduced), while the same word is blocked in posts and "fuck" is blocked in comments — a word/context-specific gap. | Critical | Fail |
| TXT-03 | Leetspeak obfuscation is still caught | Leetspeak variant of a banned word was correctly rejected. | High | Pass |

| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| TXT-04 | Case variations are caught | All-caps variant correctly rejected. | Medium | Pass |
| TXT-05 | Legit words with an embedded banned substring aren't flagged | "Scunthorpe" posted successfully — no over-flagging. | High | Pass |
| TXT-06 | Context-aware analysis allows benign use of an ambiguous word | Not run this session. | Medium | Not run |
| TXT-07 | Posts over 500 characters are rejected | 501-char post correctly rejected server-side. Note: the UI's own textarea caps at 280, well below the server's real 500-char limit. | High | Pass |
| TXT-08 | Comments over 280 characters are rejected | 281-char comment correctly rejected. | High | Pass |
| TXT-09 | Empty / whitespace-only content is rejected | All three cases (blank post, empty comment, null comment) correctly rejected. | High | Pass |
| TXT-10 | DMs aren't run through the profanity filter | Not run — skipped by scope decision; source confirms messages/route.js has no filter check. | Medium | Not run |

## API Request Runtime

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| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| PERF-01 | GET /api/posts response time under load | Couldn't time the real path — unauthenticated calls 500 (see NEW-01); only the error path was timed (~1s). | High | Blocked |
| PERF-02 | GET /api/search response time | Returned well under the 800ms target (small production dataset). | Medium | Pass |
| PERF-03 | POST /api/posts with image (Vision round-trip) | Full Vision + Storage + DB round trip completed in 2.2s for a small test image. | High | Pass |

| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| PERF-04 | Concurrent identical requests shouldn't duplicate posts | Two simultaneous identical POSTs both succeeded — no server-side idempotency, only a client-side disabled-button guard. | Medium | Fail |
| PERF-05 | Behavior when Vision API is slow/unreachable | Can't block outbound calls to Vision from the client side; not exercised. | High | Not run |
| PERF-06 | Deep-page pagination performance | Page 1 vs. page 20 differed by ~45ms; small dataset doesn't stress-test OFFSET pagination at real scale. | Low | Pass |
| PERF-07 | Realtime subscriptions don't leak connections | No errors/degradation observed across many opened/closed channels, but true verification needs Supabase dashboard access. | Medium | Blocked |
| PERF-08 | API responses set sensible cache headers | Static-ish endpoints (categories, campuses) return max-age=0 — a missed caching opportunity, not a bug. | Low | Pass |

## Direct Messages

8 / 8

| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| MSG-01 | Send a message in an existing conversation | Created and confirmed in the thread. | High | Pass |
| MSG-02 | Can't read a conversation you're not part of | Correctly returned 403 for a non-member request. | Critical | Pass |
| MSG-03 | Can't send into a conversation you're not part of | The POST handler has no participant check at all (unlike GET) — confirmed by source review; this test's ID was only caught by a foreign-key error, not an access check. | Critical | Fail |
| MSG-04 | Messages over 2000 characters are rejected | Correctly rejected before sending. | Medium | Pass |

| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| MSG-05 | Empty / whitespace-only message is rejected | Correctly rejected, nothing sent. | Medium | Pass |
| MSG-06 | New message appears live for the other participant | Appeared within the watch window via the DM realtime channel, no refresh needed. | High | Pass |
| MSG-07 | Start a new conversation with another user | New conversation created via get_or_create_dm; confirmed idempotent (repeat call returns the same conversation). | High | Pass |
| MSG-08 | Conversation list only shows the user's own conversations | Spot-checked correctly scoped. | Critical | Pass |

## Mobile vs Desktop Browser

8 / 8

| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| RESP-01 | Feed layout on mobile viewport | No horizontal overflow; mobile bottom nav renders cleanly. | High | Pass |
| RESP-02 | Feed layout on tablet viewport | No overflow, sensible layout. | Medium | Pass |
| RESP-03 | Desktop layout at wide viewport | Renders correctly at 1920×1080; surfaced an uncatalogued "Trending on Howler" panel. | Medium | Pass |
| RESP-04 | Comments modal usability on mobile | Scrollable and dismissible via the close button, but Escape does not close it (same root cause as A11Y-04). | High | Fail |
| RESP-05 | Image upload flow on mobile | File picker, preview, and submit all worked correctly on a mobile viewport. | Medium | Pass |
| RESP-06 | Messages panel usability on small screens | Clean stacked list→thread navigation with a back button, no layout breakage. | Medium | Pass |
| RESP-07 | Touch interactions work without a mouse | Like/comment taps worked identically to clicks; no hover-only functionality found. | High | Pass |

| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| RESP-08 | Orientation change doesn't break layout | No overflow after rotating to landscape. | Low | Pass |

## Search

5 / 5

| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| SRCH-01 | Search posts by partial content | Correct substring matches returned. | High | Pass |
| SRCH-02 | Search profiles by username/name | Correct matches returned. | High | Pass |
| SRCH-03 | Empty search query | Returned empty arrays without querying the DB. | Medium | Pass |
| SRCH-04 | Wildcard/injection-style queries are sanitized | SQLi payload safely neutralized — but a literal '%' query returns a broad "match everything" result since input isn't escaped before entering the ilike pattern. | High | Fail |
| SRCH-05 | Search results exclude sensitive fields | Only id/username/full_name/avatar_url exposed. | Medium | Pass |

## Likes

4 / 4

| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| LIKE-01 | Like a post for the first time | Count went 0→1 via the real UI tap, confirmed server-side. | High | Pass |
| LIKE-02 | Unlike (toggle off) | Count correctly reverted 1→0. | High | Pass |
| LIKE-03 | Rapid double-tap doesn't desync the count | Two concurrent requests resolved deterministically — one liked, one unliked, net change zero. | Medium | Pass |
| LIKE-04 | Like a non-existent post_id | Unhandled 500 with a raw foreign-key constraint error. | Low | Fail |

## Authentication

8 / 8

| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| AUTH-01 | Google OAuth login with a UW account | Observed directly: real UW Google login redirected correctly, session cookie set, username auto-generated from email. | Critical | Pass |
| AUTH-02 | Microsoft OAuth login (UW Azure tenant) | Not exercised this session (logged in via Google); JWT shows a linked, previously-used UW Azure identity as indirect evidence. | Critical | Blocked |
| AUTH-03 | Non-@uw.edu Google account should be rejected | Needs a non-UW Google account. LoginButton only sends a client-side hd:'uw.edu' hint — not confirmed enforced server-side. | Critical | Not run |
| AUTH-04 | GET /api/auth with no session | Correctly returned 401 {authenticated:false}. | High | Pass |
| AUTH-05 | Session persists across reload / new tab | Stayed authenticated after reload and in a fresh tab. | High | Pass |
| AUTH-06 | Logout clears session fully | Logout returned session to logged-out state immediately; run last since it invalidates the test session. | Critical | Pass |
| AUTH-07 | Repeat login doesn't duplicate the profile row | Profile stayed consistent across two logins; not directly row-counted in the DB. | High | Pass |
| AUTH-08 | Unauthenticated user can't call mutating routes | All 7 mutating endpoints correctly returned 401 with no side effects. | Critical | Pass |

## Rate Limiting

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| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| RATE-01 | General reads capped at 60/min | Burst of 65 unauthenticated GETs: succeeded until the shared window filled, then correctly 429'd. | High | Pass |
| RATE-02 | Writes capped at 20/min | 22-request burst: rejected exactly the requests over the 20/min | High | Pass |

| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| | | sliding window. | | |
| RATE-03 | Rate limit is scoped per-user, not global | Source-confirmed (key is user:{id}); RATE-02's exact counts are indirect behavioral confirmation. | Critical | Pass |
| RATE-04 | Falls back to IP for anonymous requests | Anonymous burst was throttled per-IP with correct headers. | Medium | Pass |
| RATE-05 | Sliding window resets after cooldown | Confirmed via the image-upload limiter's real 3-minute reset; general limiter reset is source-confirmed. | Medium | Pass |
| RATE-06 | Image uploads capped at 3 per 3 minutes | 3rd upload succeeded, 4th correctly blocked with 429. | High | Pass |
| RATE-07 | Image upload throttle resets after 3 minutes | 3 further uploads succeeded after waiting out the window. | Medium | Pass |
| RATE-08 | 429 responses include backoff headers | X-RateLimit-Limit/Remaining/Reset all present and accurate. | Low | Pass |

## Profile Management

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| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| PROF-01 | View own profile page | Correct data returned. | High | Pass |
| PROF-02 | View another user's public profile | Correct profile + posts returned. | High | Pass |
| PROF-03 | Non-existent username | Clean 404, no crash. | Medium | Pass |
| PROF-04 | Edit profile details | Bio update persisted and reverted cleanly. | High | Pass |
| PROF-05 | Username collision handling | Not testable without a second colliding-email account; source shows no uniqueness/suffix logic on the derived username. | Medium | Blocked |

| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| PROF-06 | My-profile likes list is accurate | Correctly reflected a controlled like/unlike. | Medium | Pass |

## Cross-Browser Compatibility

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| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| BROW-01 | Core flows in Chrome | Feed loads cleanly, zero console errors. | High | Pass |
| BROW-02 | Core flows in Safari (WebKit engine) | Clean in WebKit, though this isn't identical to real Safari's ITP/cookie behavior — a strong signal, not a substitute. | High | Pass |
| BROW-03 | Core flows in Firefox | Clean, aside from a benign Cloudflare cookie-scope console warning from Supabase's storage CDN (not a Howler bug). | Medium | Pass |
| BROW-04 | Core flows in Edge | Playwright has no real Edge engine; rendering is likely fine (Chromium-based) but the Microsoft/Azure SSO interplay wasn't verified. | Medium | Blocked |
| BROW-05 | Graceful degradation on a throttled connection | Shows a proper loading state rather than a blank screen, though the full feed took ~42s under heavy throttling. | Medium | Pass |

## Categories & Campus

4 / 4

| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| CAT-01 | Category list loads for post creation | 11 categories returned in 0.42s. One data-hygiene issue: "Pictures/Art" has a stray leading newline in its stored name (not visibly rendered). | Medium | Pass |
| CAT-02 | Category filter chip filters the feed | Clicking a chip correctly re-fetches and re-renders the filtered feed. | Medium | Pass |

| ID | TEST CASE | RESULT | PRIORITY | STATUS |
|---|---|---|---|---|
| CAT-03 | Campus list loads correctly | 3 campuses returned correctly. | Low | Pass |
| CAT-04 | Profile shows the correct campus name via join | Resolved correctly to "UW Seattle". | Low | Pass |

Howler QA Audit — 135 test cases across 19 categories, executed against production on 2026-07-18. Prepared for internal engineering review.